

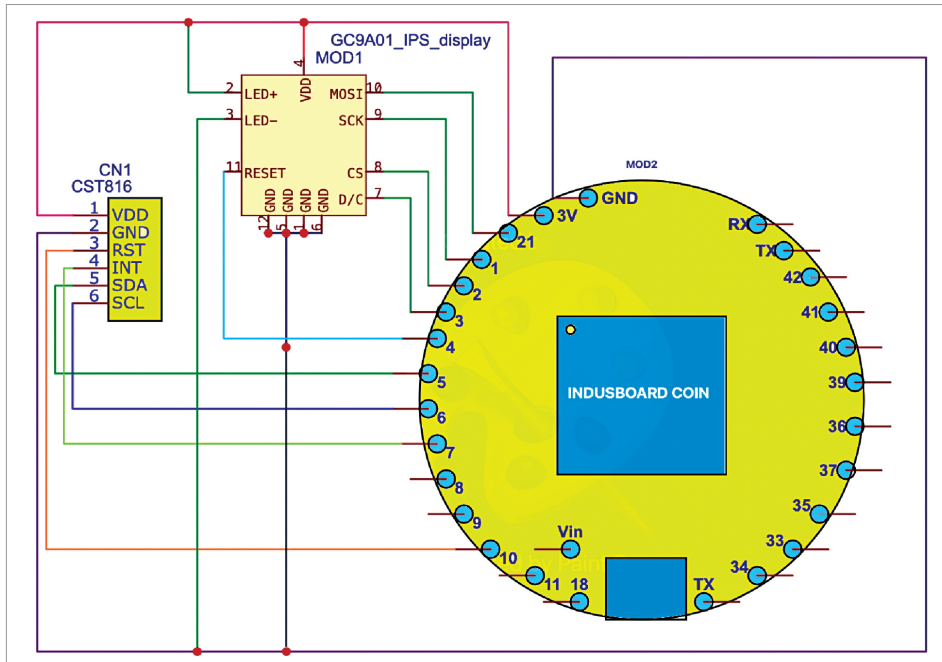


Fig. 3: Circuit diagram

SSID. It is built around the Indusboard coin type, round touch display GC9A01 driver (MOD1), and touch sensor CST816 (CN1). A USB Type C is used to connect the Indusboard to a laptop or desktop.

The connections are straightforward. Connect pins 21, 1, 2, 3, and 4 of the Indusboard to the MOSI, SCK, CS, D/C, and RESET pins of the GC9A01 driver (MOD1), respectively. Similarly, connect pins 5, 6, 7, and 9 of the Indusboard to the SDA, SCL, INT, and RESET pins of the CST816 (CN1), respectively.

Construction and testing

First, configure the Wi-Fi network to connect with and extract the location, time zone, and other data. After completing the configuration in the source code, upload the source code by selecting the Indusboard or ESP32S2 and the COM port.

Next, solder the components as shown in Fig. 4. After proper assembly, your device is ready for testing.

Testing of the SSID-based location tracking system involves the following steps:

Power the device using the USB-C or 3.3V battery connected to the 3V and GND pins on the board. Wait a few seconds; it will display a “connecting” message. After a successful connection, it will display data such as country, population, time zone, country code, latitude, longitude, city, state, pin code, network name, IP address, and much more. The final data displayed on the device is shown in Fig. 5. **EFY**

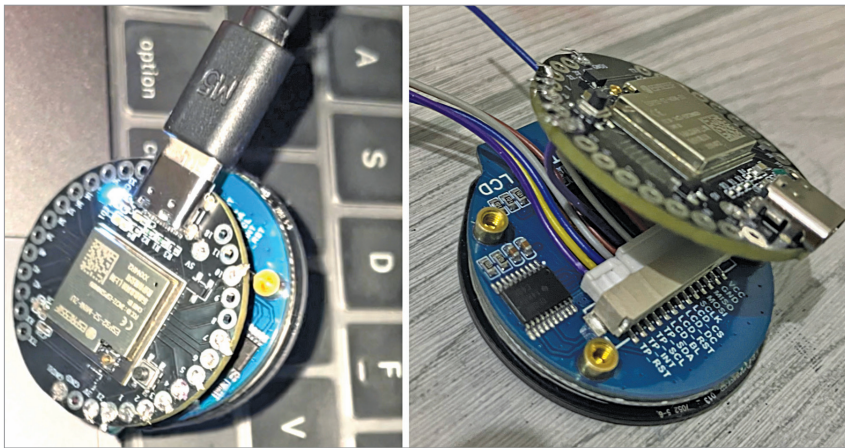


Fig. 4: Component soldering



Fig. 5: Displaying the data

EFY Note

The source code of this project is available for free download at www.electronicshobby.com

Ashwini Kumar Sinha, an IoT and AI enthusiast, is a tech journalist at EFY